CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

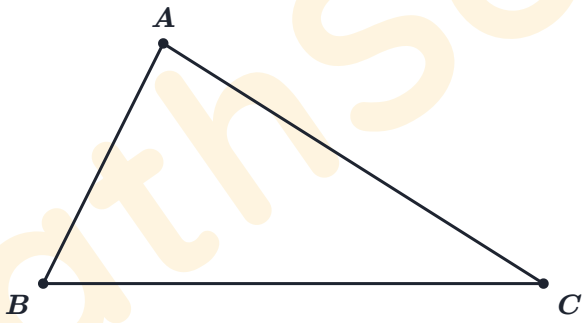
Chapter 11: Inequalities

Topic-wise short notes with solved examples

Inequalities in a triangle: the greater side has the greater angle opposite it and conversely, the sum of two sides exceeds the third, and the perpendicular is the shortest distance from a point to a line.

1 Side and angle

- **Theorem 1**: if two sides of a triangle are unequal, the **greater side has the greater angle opposite to it**.
- **Theorem 2 (converse)**: if two angles of a triangle are unequal, the **greater angle has the greater side opposite to it**.
- So the largest angle is opposite the longest side, and the smallest angle is opposite the shortest side.



BC is the longest side, so angle A is the largest angle

- In a **right-angled triangle** the hypotenuse is the longest side, because the right angle is the largest angle.
- In an **obtuse-angled triangle** the side opposite the obtuse angle is the longest.

Example 1.1 In triangle ABC, $AB = 5$ cm, $BC = 8$ cm and $CA = 7$ cm. Arrange the angles in ascending order.

SOLUTION

$\angle C$ is opposite AB, $\angle B$ is opposite CA, $\angle A$ is opposite BC.

$\angle C < \angle B < \angle A$

Example 1.2 In triangle PQR, $\angle P = 70^\circ$ and $\angle Q = 50^\circ$. Arrange the sides in descending order.

SOLUTION

$\angle R = 60^\circ$. The sides opposite $70^\circ, 60^\circ, 50^\circ$ are QR, PQ, PR.

$QR > PQ > PR$

2 Triangle inequality

- **Theorem 3**: the **sum of any two sides** of a triangle is **greater than the third side**.
- The **difference of any two sides** is **less than the third side**.
- Three lengths $a \leq b \leq c$ form a triangle only when $a + b > c$.
- If two sides are a and b (with $a > b$), the third side x satisfies $a - b < x < a + b$.

Example 2.1 Can a triangle have sides 4 cm, 5 cm and 10 cm?

SOLUTION

$4 + 5 = 9$, which is less than 10.

No

Example 2.2 Two sides of a triangle are 6 cm and 9 cm. Between what limits does the third side lie?

SOLUTION

$9 - 6 < x < 9 + 6$

Between 3 cm and 15 cm

Example 2.3 Two sides of a triangle are 5 cm and 11 cm and the third side is a whole number of centimetres. How many triangles are possible?

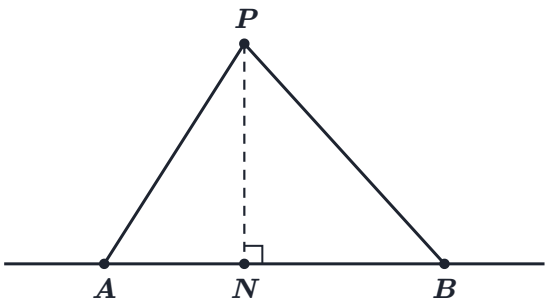
SOLUTION

$6 < x < 16$, so x can be 7, 8, ..., 15.

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3 Shortest distance

- **Theorem 4**: of all the line segments that can be drawn from a point to a line, the **perpendicular is the shortest**.
- The length of this perpendicular is the **distance of the point from the line**.



PN is perpendicular to the line, so PN is shorter than PA and PB

- Reason: in right triangle PNA, the right angle at N is the largest angle, so the hypotenuse PA is the longest side.

4 Proving inequalities

- Use the exterior angle property: an exterior angle is **greater** than each interior opposite angle.
- Use isosceles triangles to move a known equality into the triangle you need.
- Apply the triangle inequality to each triangle in the figure and add the results.

Example 4.1 In triangle ABC, $AB > AC$. D is a point on BC. Prove that $AB > AD$.

SOLUTION

$AB > AC$ gives $\angle C > \angle B$.

$\angle ADB$ is an exterior angle of $\triangle ADC$, so $\angle ADB > \angle C > \angle B$.

In $\triangle ABD$, $\angle ADB > \angle B$, so $AB > AD$

Example 4.2 D is a point on side BC of triangle ABC such that $AD = AC$. Prove that $AB > AD$.

SOLUTION

$AD = AC$ gives $\angle ADC = \angle ACD$.

$\angle ADC$ is an exterior angle of $\triangle ABD$, so $\angle ADC > \angle B$. Hence $\angle ACB > \angle B$.

So $AB > AC$, and since $AC = AD$, $AB > AD$

Example 4.3 Prove that the perimeter of a triangle is greater than the sum of its three medians.

SOLUTION

Let AD be a median. Produce AD to E with $DE = AD$. Then $\triangle ADB \cong \triangle EDC$, so $CE = AB$.

In $\triangle ACE$: $AC + CE > AE$, that is $AC + AB > 2AD$. Similarly $AB + BC > 2BE$ and $BC + CA > 2CF$.

Adding and dividing by 2: $AB + BC + CA > AD + BE + CF$

Example 4.4 In quadrilateral ABCD, prove that $AB + BC + CD + DA > AC + BD$.

SOLUTION

From the four triangles: $AB + BC > AC$, $CD + DA > AC$, $AB + AD > BD$, $BC + CD > BD$.

Adding gives $2(AB + BC + CD + DA) > 2(AC + BD)$

Example 4.5 O is any point inside triangle ABC. Prove that $OA + OB + OC > \frac{1}{2}(AB + BC + CA)$.

SOLUTION

$OA + OB > AB, OB + OC > BC, OC + OA > CA$

Adding: $2(OA + OB + OC) > AB + BC + CA$

5 Common mistakes

- Matching a side with an adjacent angle instead of the angle **opposite** it.
- Checking the triangle inequality for only one pair when the pair chosen is not the two shortest sides.
- Writing "sum of two sides is equal to or greater than the third". It must be strictly greater.
- Saying an exterior angle equals an interior opposite angle. It equals their **sum**, so it is greater than each.
- Forgetting the lower limit for the third side.

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